Skill Builder 5 Pivot Tables & Pivot Charts

In this activity, you will learn how to summarise data using Pivot Tables and present it clearly using Pivot Charts.

You will move from:

- raw data
- to structured summaries
- to visual insights

What you will use

- Pivot Table
- Pivot Chart

Key Idea

We are now answering:

“What patterns exist in this data?”

Instead of counting rows one at a time, Excel will:

- group data
- summarise it
- display results clearly

How this builds from previous learning

- COUNT → total
- COUNTIF → one rule
- COUNTIFS → multiple rules
- Pivot Table → **all of this, but automatic and flexible**

SECTION 1 — What is a Pivot Table

A Pivot Table turns a large dataset into a simple summary table.

It helps you quickly answer questions like:

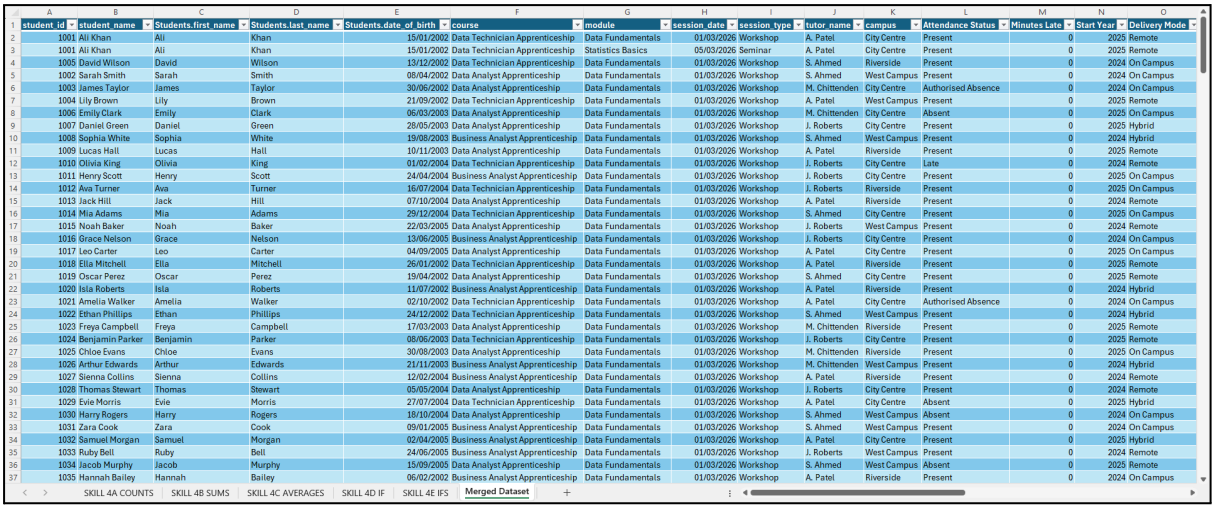
- How many Present vs Absent?
- Which category is highest?
- Where are the trends?

What Excel is doing

Excel:

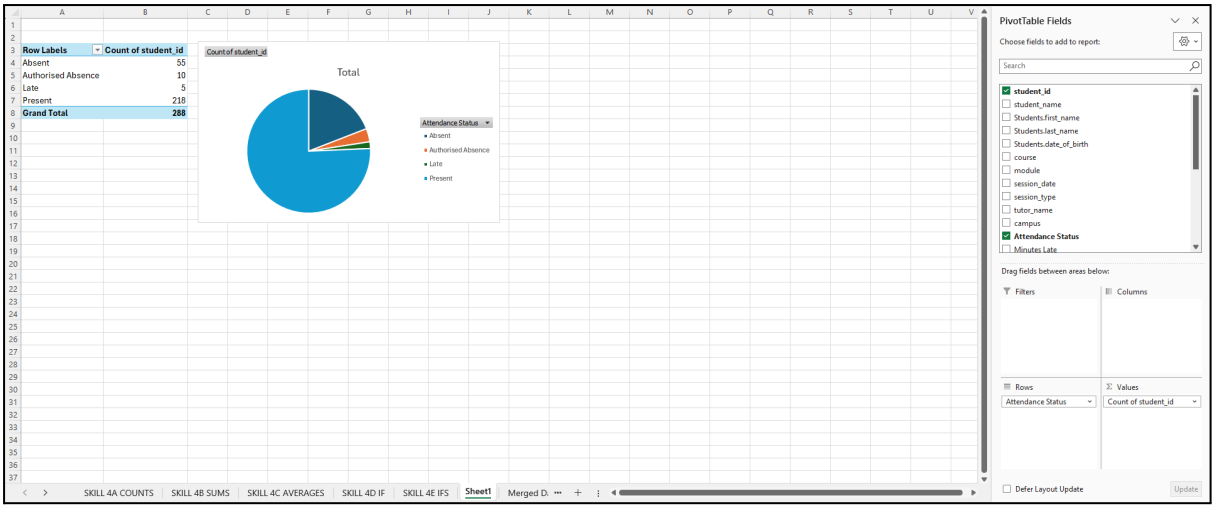
- reads every row
- groups matching values
- calculates totals (usually COUNT)
- displays the result

Visual Placeholder



student_id	student_name	Students.first_name	Students.last_name	Students.date_of_birth	course	module	session_date	session_type	tutor_name	campus	Attendance Status	Minutes Late	Start Year	Delivery Mode
1001	Ali Khan	Ali	Khan	15/01/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Present	0	2025	Remote
1001	Ali Khan	Ali	Khan	15/01/2002	Data Technician Apprenticeship	Statistics Basics	05/03/2026	Seminar	A. Patel	City Centre	Present	0	2025	Remote
1005	David Wilson	David	Wilson	13/12/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	Riverside	Present	0	2024	On Campus
1002	Sarah Smith	Sarah	Smith	08/04/2002	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0	2024	On Campus
1003	James Taylor	James	Taylor	30/06/2002	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	City Centre	Authorised Absence	0	2024	On Campus
1004	Lily Brown	Lily	Brown	21/09/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	West Campus	Present	0	2025	Remote
1006	Emily Clark	Emily	Clark	06/03/2003	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	City Centre	Absent	0	2025	On Campus
1007	Daniel Green	Daniel	Green	28/05/2003	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0	2025	Hybrid
1008	Sophia White	Sophia	White	19/08/2003	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0	2024	Hybrid
1009	Lucas Hall	Lucas	Hall	10/11/2003	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2025	Remote
1010	Olivia King	Olivia	King	01/02/2004	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Late	0	2024	Remote
1011	Henry Scott	Henry	Scott	24/04/2004	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0	2025	On Campus
1012	Ava Turner	Ava	Turner	16/07/2004	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	Riverside	Present	0	2025	On Campus
1013	Jack Hill	Jack	Hill	07/10/2004	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2024	Remote
1014	Mia Adams	Mia	Adams	29/12/2004	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	City Centre	Present	0	2025	On Campus
1015	Noah Baker	Noah	Baker	22/03/2005	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	West Campus	Present	0	2024	Remote
1016	Grace Nelson	Grace	Nelson	13/06/2005	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0	2024	On Campus
1017	Leo Carter	Leo	Carter	04/09/2005	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Present	0	2025	On Campus
1018	Ella Mitchell	Ella	Mitchell	26/01/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2025	Remote
1019	Oscar Perez	Oscar	Perez	19/04/2002	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	City Centre	Present	0	2025	Remote
1020	Isla Roberts	Isla	Roberts	11/07/2002	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2024	Hybrid
1021	Amelia Walker	Amelia	Walker	02/10/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Authorised Absence	0	2024	On Campus
1022	Ethan Phillips	Ethan	Phillips	24/12/2002	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0	2024	Hybrid
1023	Freya Campbell	Freya	Campbell	17/03/2003	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	Riverside	Present	0	2025	Remote
1024	Benjamin Parker	Benjamin	Parker	08/06/2003	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0	2025	Remote
1025	Chloe Evans	Chloe	Evans	30/08/2003	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	Riverside	Present	0	2025	On Campus
1026	Arthur Edwards	Arthur	Edwards	21/11/2003	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	M. Chittenden	West Campus	Present	0	2024	Hybrid
1027	Sienna Collins	Sienna	Collins	12/02/2004	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2024	Remote
1028	Thomas Stewart	Thomas	Stewart	05/05/2004	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	City Centre	Present	0	2024	Remote
1029	Evie Morris	Evie	Morris	27/07/2004	Data Technician Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Absent	0	2025	Hybrid
1030	Harry Rogers	Harry	Rogers	18/10/2004	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Absent	0	2024	On Campus
1031	Zara Cook	Zara	Cook	09/01/2005	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Present	0	2024	On Campus
1032	Samuel Morgan	Samuel	Morgan	02/04/2005	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	City Centre	Present	0	2025	Hybrid
1033	Ruby Bell	Ruby	Bell	24/06/2005	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	J. Roberts	West Campus	Present	0	2024	Remote
1034	Jacob Murphy	Jacob	Murphy	15/09/2005	Data Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	S. Ahmed	West Campus	Absent	0	2025	Remote
1035	Hannah Bailey	Hannah	Bailey	06/02/2002	Business Analyst Apprenticeship	Data Fundamentals	01/03/2026	Workshop	A. Patel	Riverside	Present	0	2024	On Campus

SKILL 4A COUNTS | SKILL 4B SUMS | SKILL 4C AVERAGES | SKILL 4D IF | SKILL 4E IFS | Merged Dataset



Important Point

Pivot Tables do not change your data.
They are just a different way of viewing it.

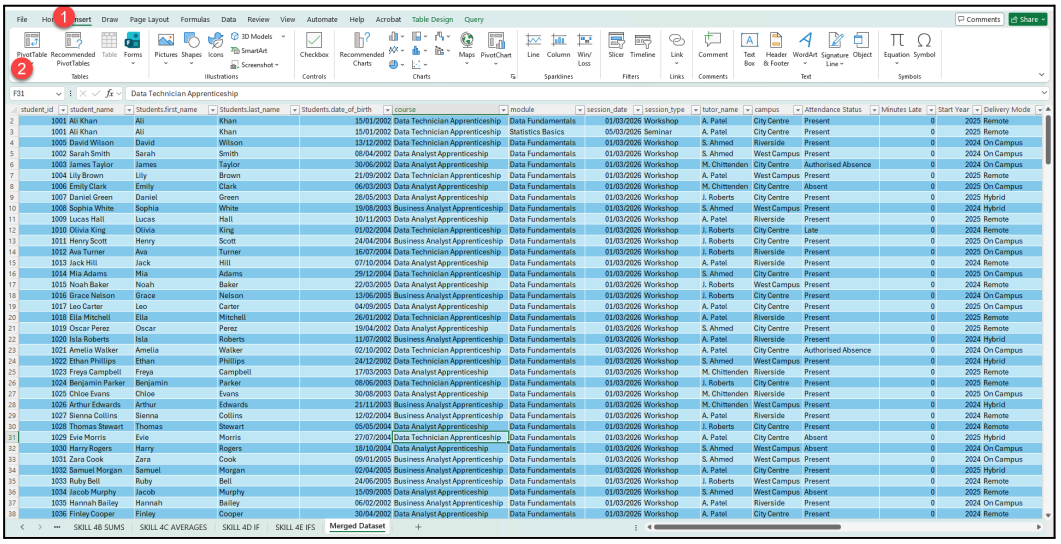
SECTION 2 — Creating a Pivot Table

Question 1

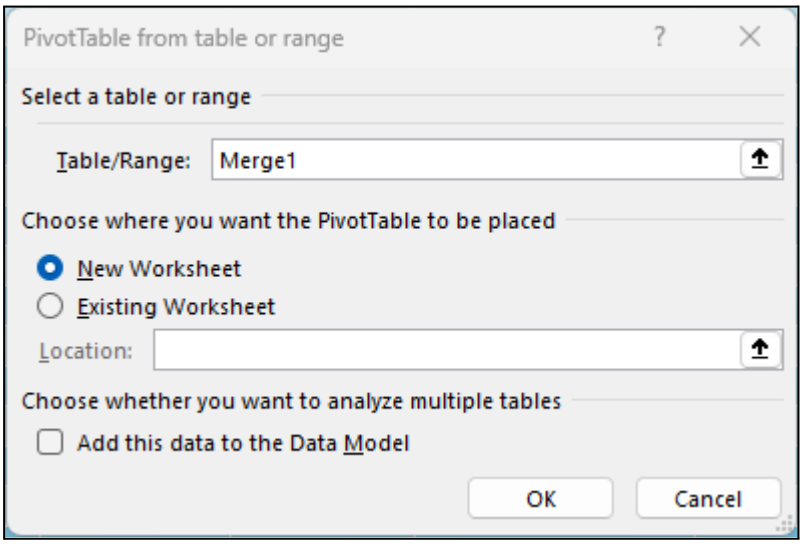
How many sessions are Present vs Absent?

What you do

1. Select your Merged Dataset table
2. Go to **Insert → Pivot Table**

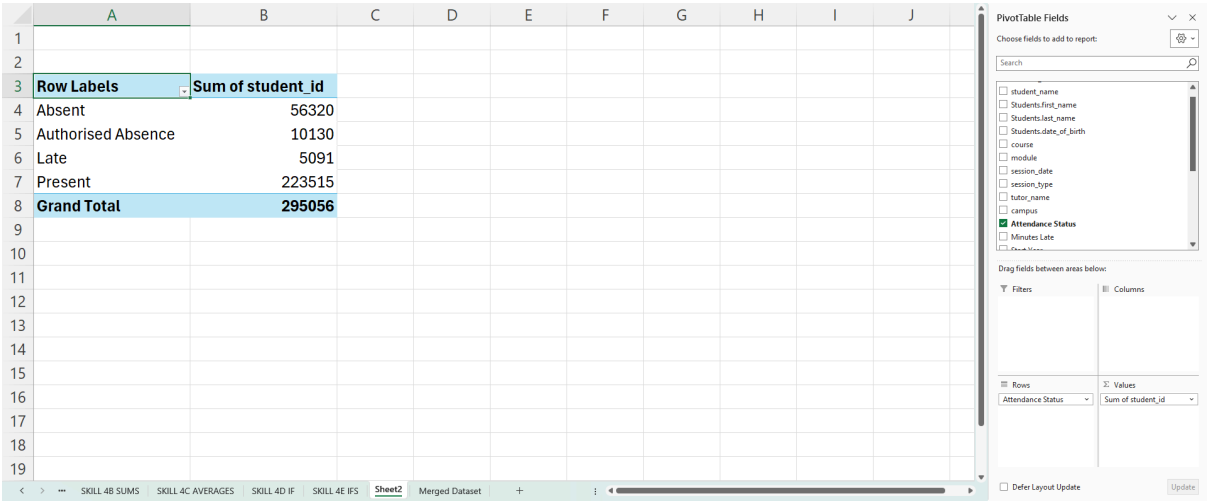


3. It should automatically select your table so click OK

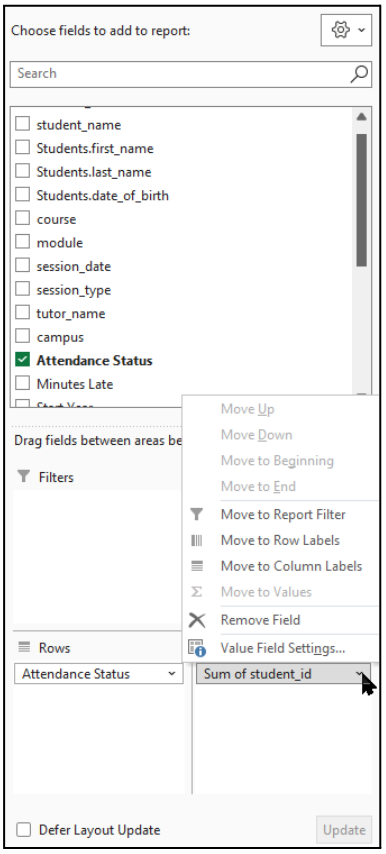


Build the table

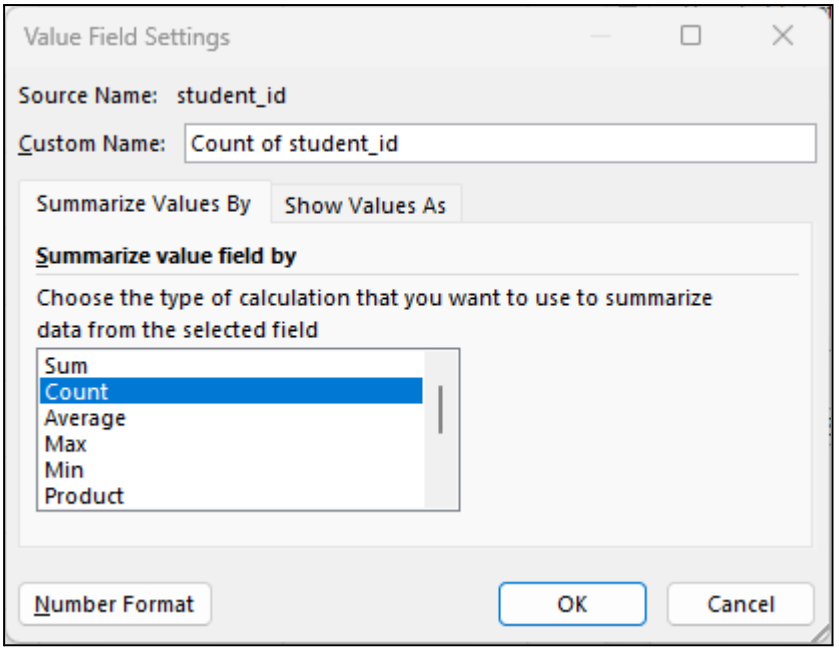
- Drag **Attendance Status→ Rows**
- Drag **Student_id→ Values**



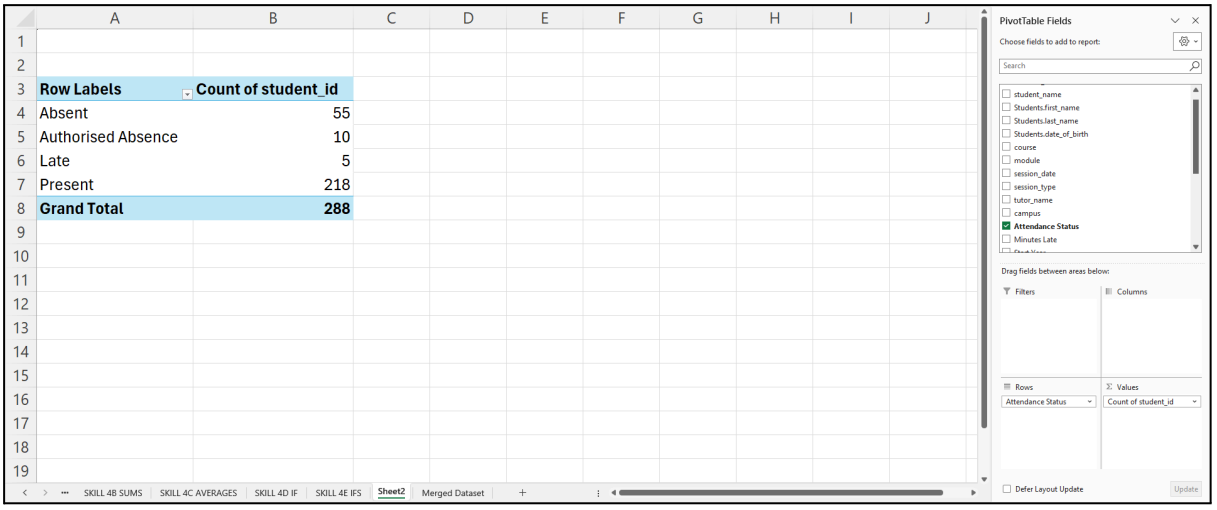
- It defaults to **sum** that we don't want so under **Values** click on the down arrow



- Change the aggregation to **Count**



- Click **OK**



What Excel is doing

For each row:

- check Attendance
- group it
- count it

Result

A simple summary table showing totals for each category

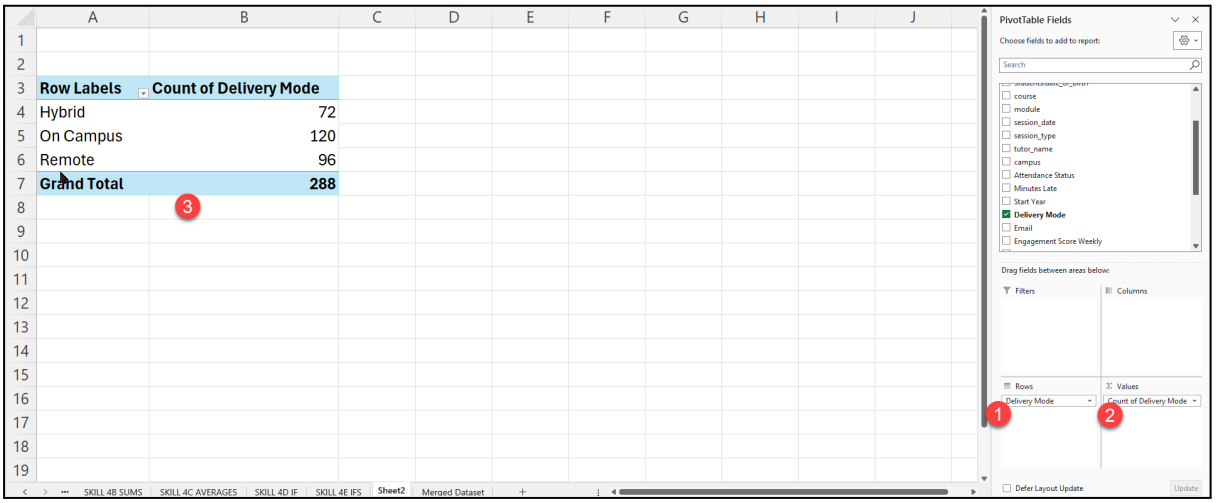
SECTION 3 — Changing the View (Pivoting)

Question 2

How many sessions per Delivery Mode?

What you change

- Remove Attendance
- Drag **Delivery Mode** → **Rows**
- Drag **Delivery Mode** → **Values**



Key Idea

You are not changing the data
You are changing the question

What Excel is doing

Same process:

- group
- count
- display

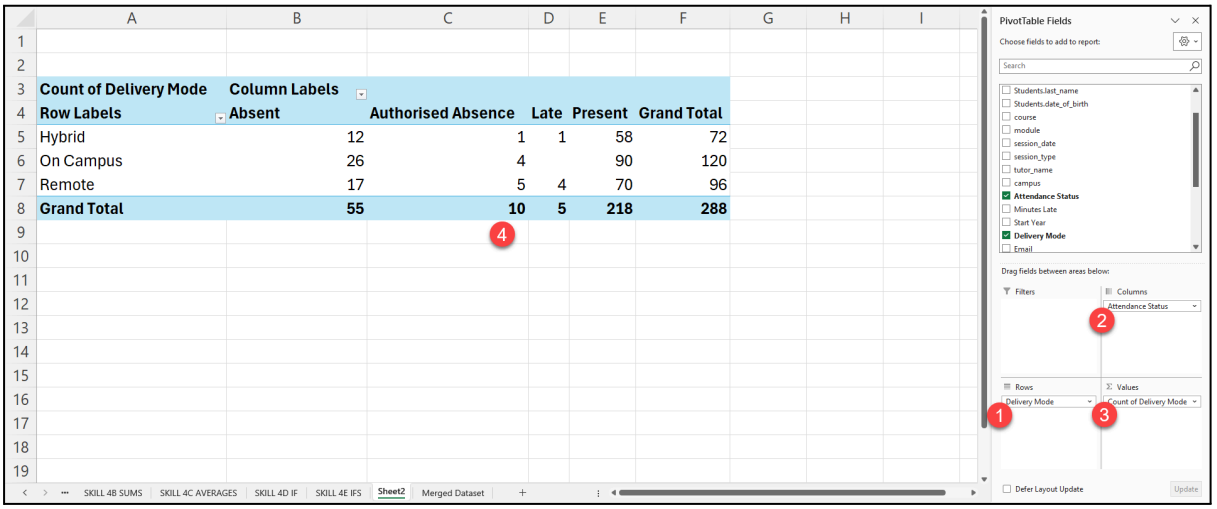
SECTION 4 — Using Multiple Fields

Question 3

How many Present and Absent sessions by Delivery Mode?

What you do

- Rows → Delivery Mode
- Columns → Attendance Status
- Values → Count



Count of Delivery Mode	Column Labels				
Row Labels	Absent	Authorised Absence	Late	Present	Grand Total
Hybrid	12	1	1	58	72
On Campus	26	4		90	120
Remote	17	5	4	70	96
Grand Total	55	10	5	218	288

What Excel is doing

For each row:

- check Delivery Mode
- check Attendance
- group both together

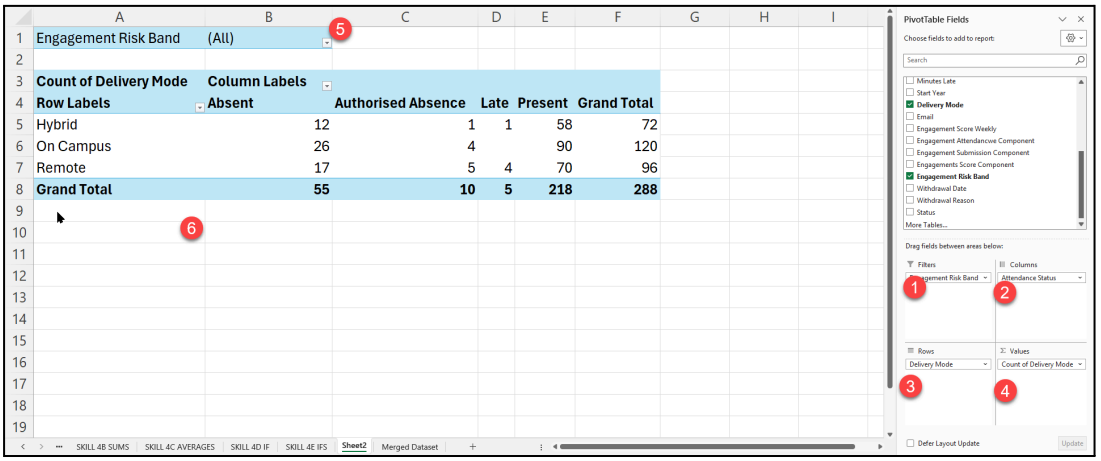
Key Idea

Pivot Tables allow comparisons across categories

SECTION 5 — Using Filters

Question 4

How many sessions for High Risk students only?

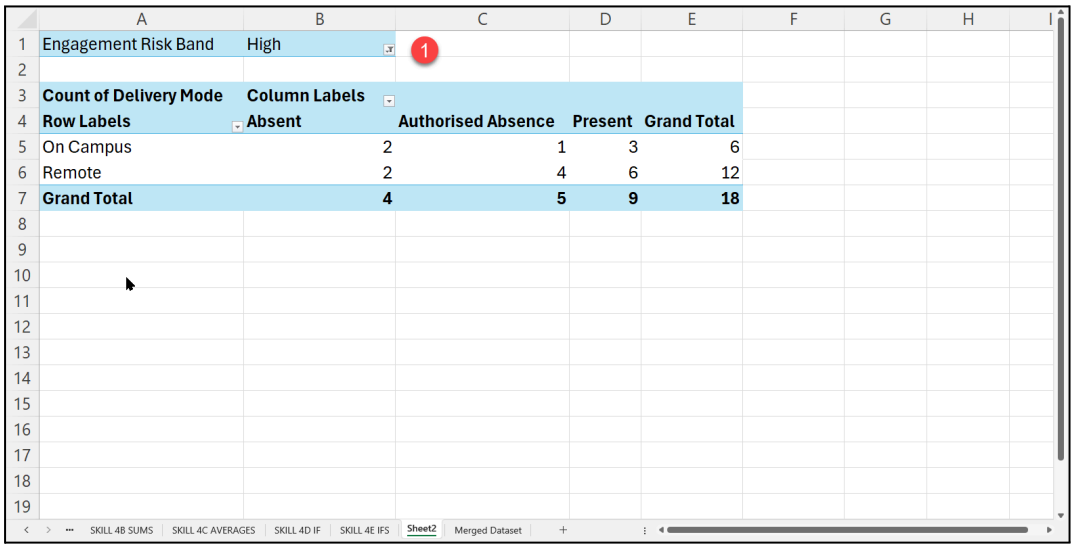


What you do

- Drag **Risk Band** → **Filters**
- Select **High**

What Excel is doing

- filters dataset first
- then builds summary



SECTION 6 — Pivot Charts (Visualising Data)

What is a Pivot Chart

A Pivot Chart is a visual representation of your Pivot Table.

It allows you to:

- quickly spot patterns
- compare values
- present findings clearly

Visual Placeholder

[Visual Placeholder – Pivot Chart (bar chart) showing Present vs Absent]

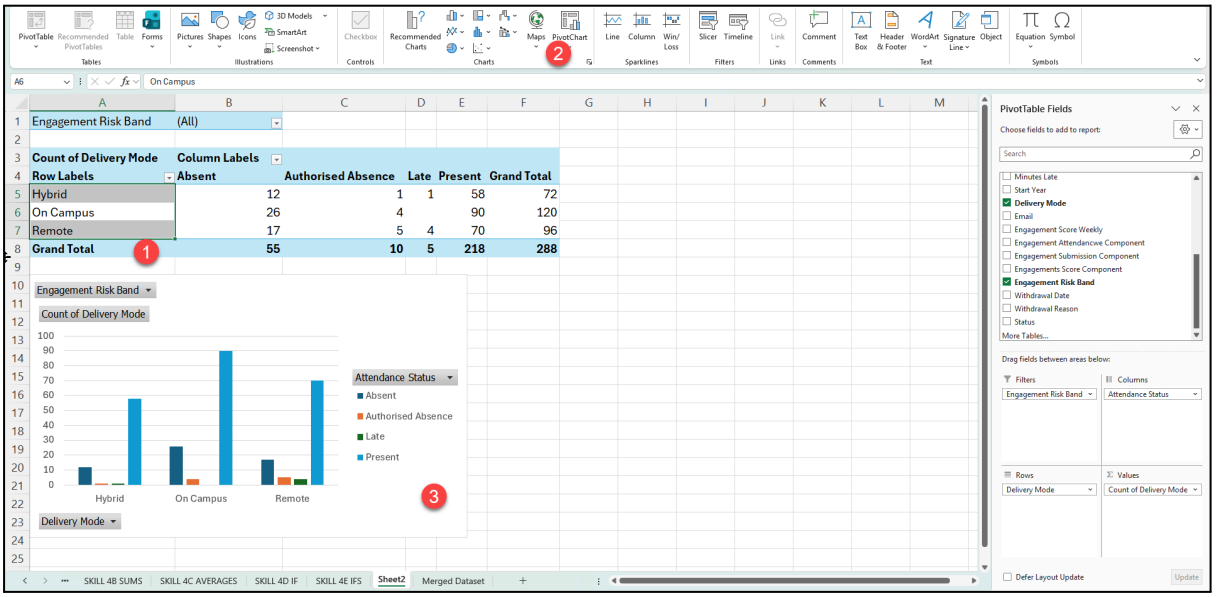
Creating a Pivot Chart

Question 5

Show Attendance as a chart

What you do

1. Click inside your Pivot Table
2. Go to **Insert** → **Pivot Chart**
3. Choose a chart type (e.g. bar chart)



What Excel is doing

Excel:

- takes your Pivot Table
- converts it into a visual format

Important Behaviour

If you change the Pivot Table, the chart updates automatically

SECTION 7 — Customising Pivot Charts

Key Idea

Pivot Charts are linked to Pivot Tables.

You customise them in TWO ways:

1. Change the Pivot Table
2. Change the chart design

Customisation 1 — Changing the Data (Pivot Table Control)

You can:

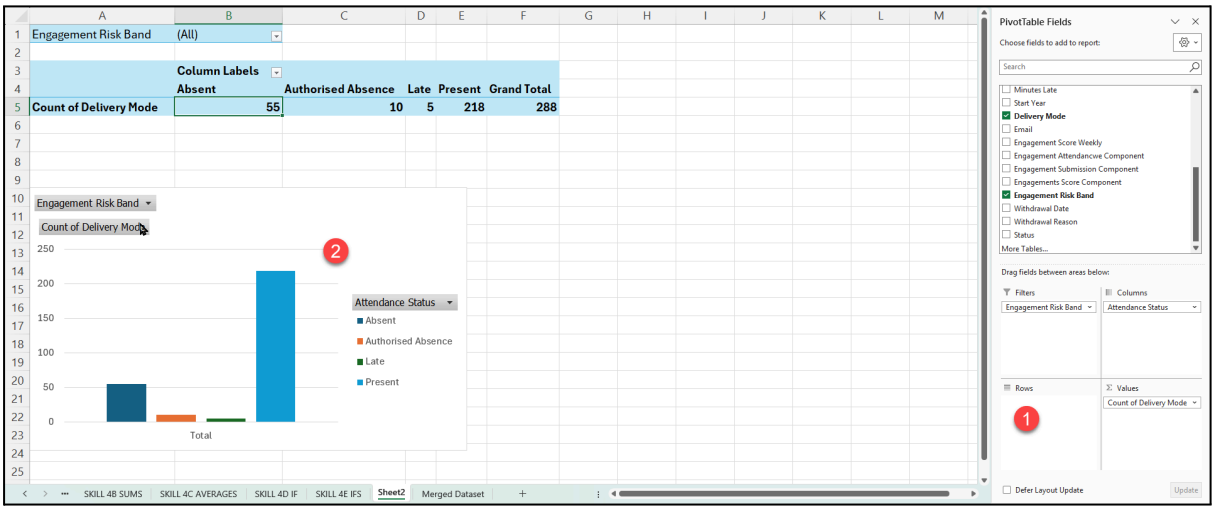
- add/remove fields
- change rows and columns
- apply filters

Example

If you change:

- Rows from Attendance → Delivery Mode

The chart will automatically update



Customisation 2 — Changing Chart Type

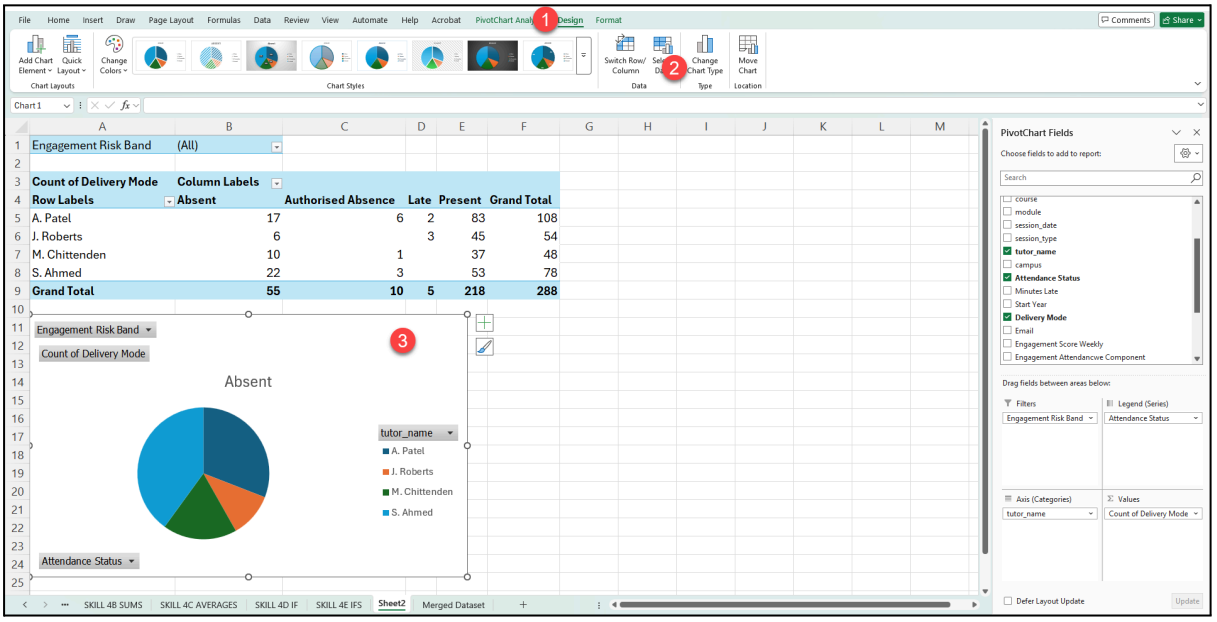
You can change how the data is shown:

- Bar chart → good for comparison

- Pie chart → good for proportions
- Line chart → good for trends

What you do

- Click the chart
- Go to **Chart Design → Change Chart Type**



Customisation 3 — Filtering from the Chart

Pivot Charts include filter buttons.

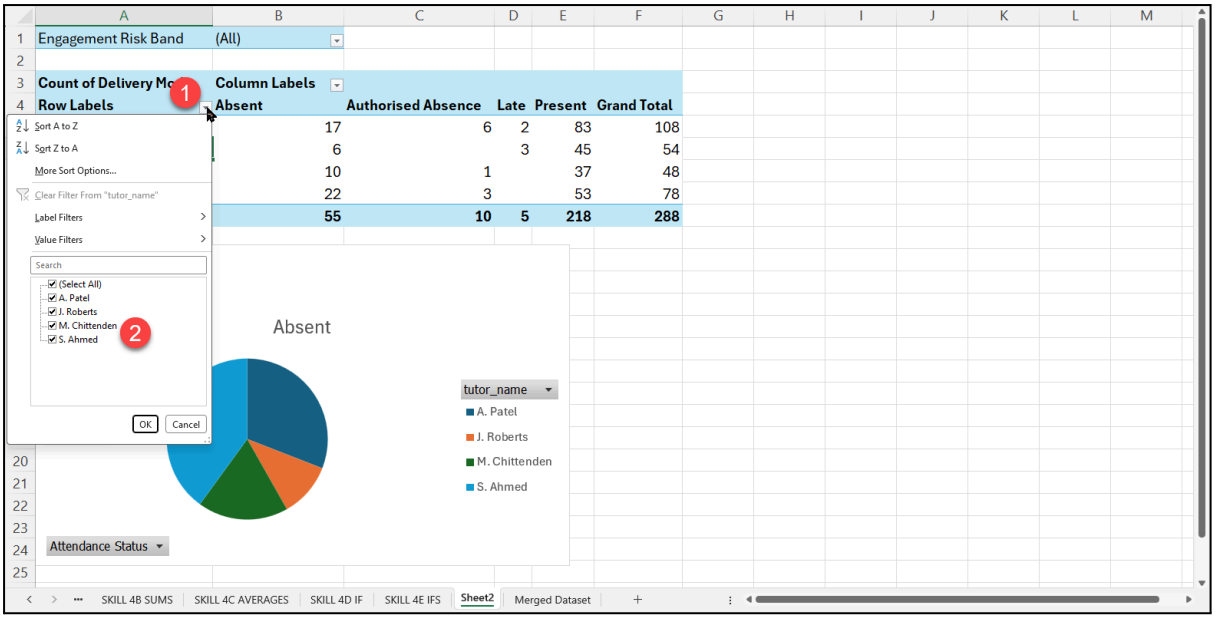
You can:

- click filters directly on the chart
- select specific categories

What Excel is doing

It updates BOTH:

- the chart
- the Pivot Table



Customisation 4 — Improving Readability

You can:

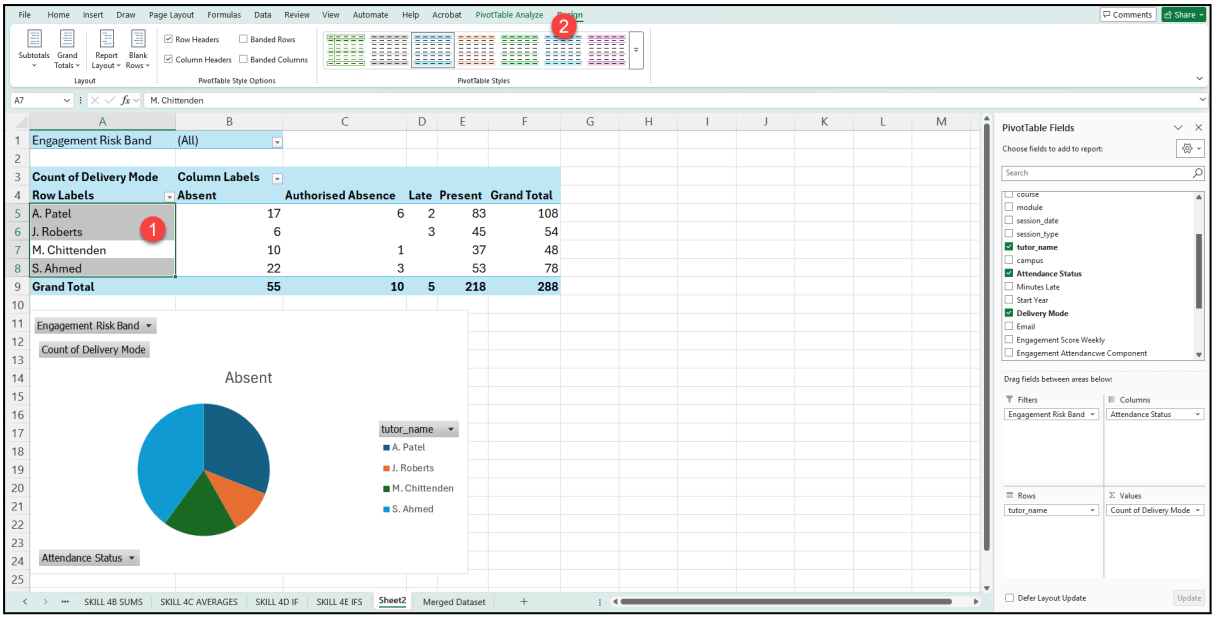
- add titles
- change colours
- adjust labels

What you do

- Click chart
- Use **Chart Design / Format tabs**

Example Improvements

- Add title: “Attendance Overview”
- Add data labels
- use clear colours


File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat PivotTable Analyze Design
Comments
Share
Subtotals
Grand Totals
Report Layout
Blank Rows
Layout
Row Headers
Banded Rows
Column Headers
Banded Columns
PivotTable Style Options
PivotTable Styles
A7
M. Chittenden
Engagement Risk Band (All)
Count of Delivery Mode Column Labels
Row Labels Absent Authorised Absence Late Present Grand Total
A. Patel 17 6 2 83 108
J. Roberts 6 3 45 54
M. Chittenden 10 1 37 48
S. Ahmed 22 3 53 78
Grand Total 55 10 5 218 288
Engagement Risk Band
Count of Delivery Mode
Absent
tutor_name
A. Patel
J. Roberts
M. Chittenden
S. Ahmed
Attendance Status
SKILL 4B SUMS
SKILL 4C AVERAGES
SKILL 4D IF
SKILL 4E IFS
Sheet2
Merged Dataset
PivotTable Fields
Choose fields to add to report:
Search
course
module
session_date
session_type
tutor_name
campus
Attendance Status
Minutes Late
Start Year
Delivery Mode
Email
Engagement Score Weekly
Engagement Attendancwe Component
Drag fields between areas below:
Filters
Engagement Risk Band
Columns
Attendance Status
Rows
tutor_name
Values
Count of Delivery Mode
Defer Layout Update
Update

SECTION 8 — Interpreting Results

Question 6

What does the chart show?

Example Explanation

- Most sessions are Present
- A smaller number are Absent
- This suggests strong attendance

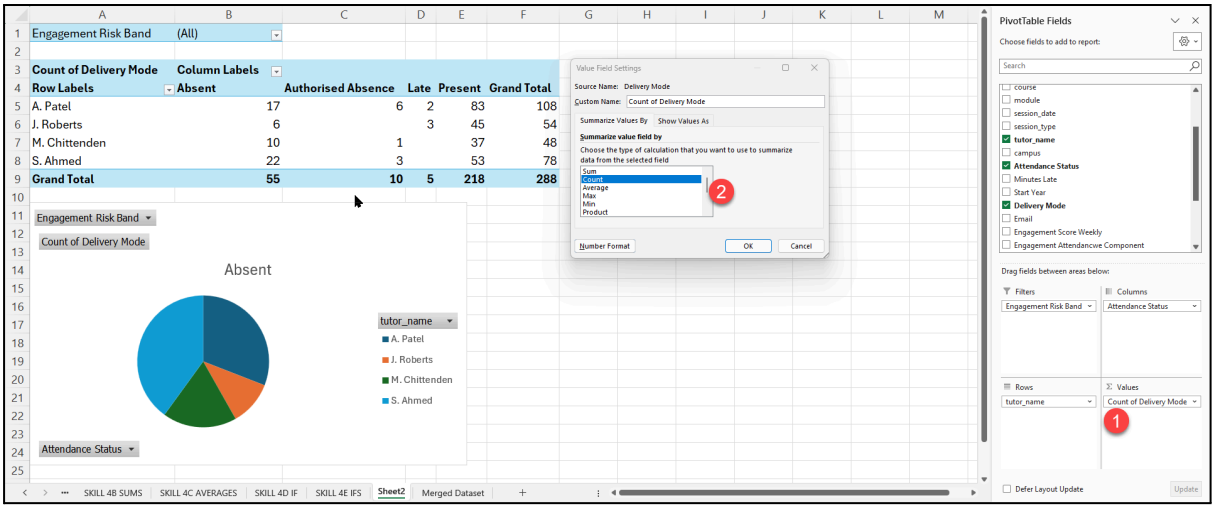
Key Skill

Analysis is not just showing data
It is explaining what it means

SECTION 9 — Common Mistakes

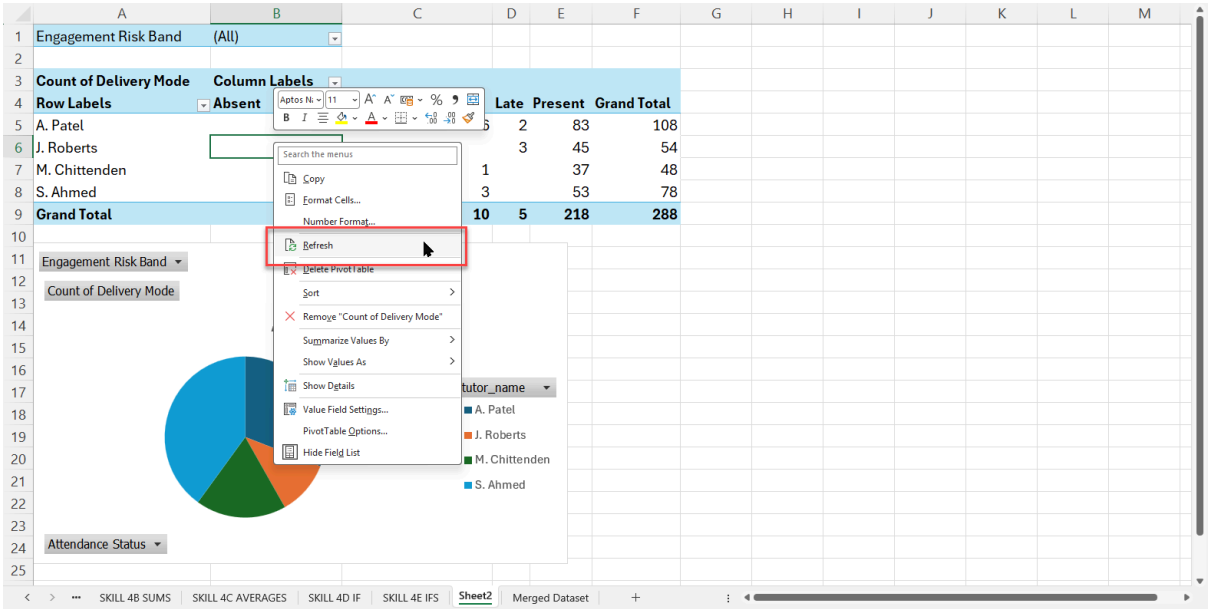
1. Wrong calculation

- If Excel shows SUM instead of COUNT
- Change Value Field Settings to COUNT



2. Data not refreshing

- If your data changes
- Right click → Refresh



3. Missing or inconsistent data

Blank or inconsistent values affect results

Final Summary

What Excel is doing every time

- reads each row
- groups data
- applies calculations
- displays results

Tool comparison

- COUNT → simple counting
- Pivot Table → structured summary
- Pivot Chart → visual insight

Simple way to remember

- Pivot Table = **“Summarise it”**
- Pivot Chart = **“Show it clearly”**

Final Note

If you get stuck, remember:

“Excel is grouping rows and helping me see the bigger picture”